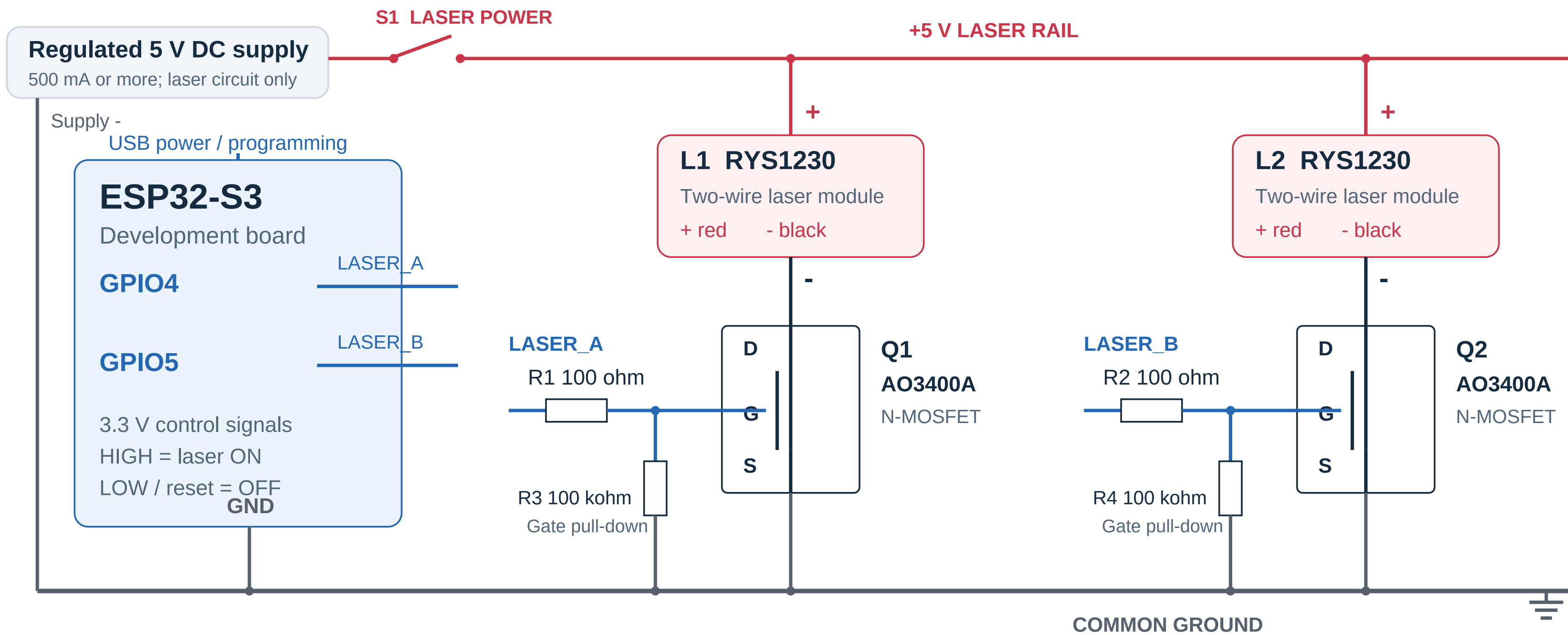


ESP32-S3 + RYS1230 laser circuit

Two independent laser channels for the ramp warning zone in Proposed Solution section 3.



WIRING RULE Connect matching LASER_A / LASER_B labels. Every ground joins the same ground rail.

Use complete 5 V-rated modules with internal current limiting. Do not connect a bare laser diode or power a laser from a GPIO.

Connections and demonstration setup

Build one channel for one module, or both channels for two modules. No motor or ramp actuator is connected here.

Connection	Destination / detail
ESP32-S3 GPIO4	R1 100 ohm -> Q1 gate (G)
ESP32-S3 GPIO5	R2 100 ohm -> Q2 gate (G)
Q1 / Q2 gate	100 kohm resistor to its source / GND
Q1 / Q2 drain (D)	Corresponding RYS1230 negative lead
Q1 / Q2 source (S)	Common GND rail
RYS1230 positive leads	S1 switched +5 V laser rail
External supply negative	ESP32 GND and both MOSFET sources
ESP32 USB	Controller power; leave its 5V / VIN pin unconnected

Optional model request button

S2: momentary, normally open



Configure GPIO13 as INPUT_PULLUP.
Press connects GPIO13 to GND.
Use a debounced press to toggle the laser demonstration; boot with lasers OFF.

Button shows the proposal's request flow.
It does not implement ramp interlocks.

Parts for the two-laser circuit

- 1x ESP32-S3 development board; 2x RYS1230 modules
- 2x AO3400A on SOT-23-to-breadboard adapters
- 2x 100 ohm; 2x 100 kohm; S1 switch; wires; 5 V supply

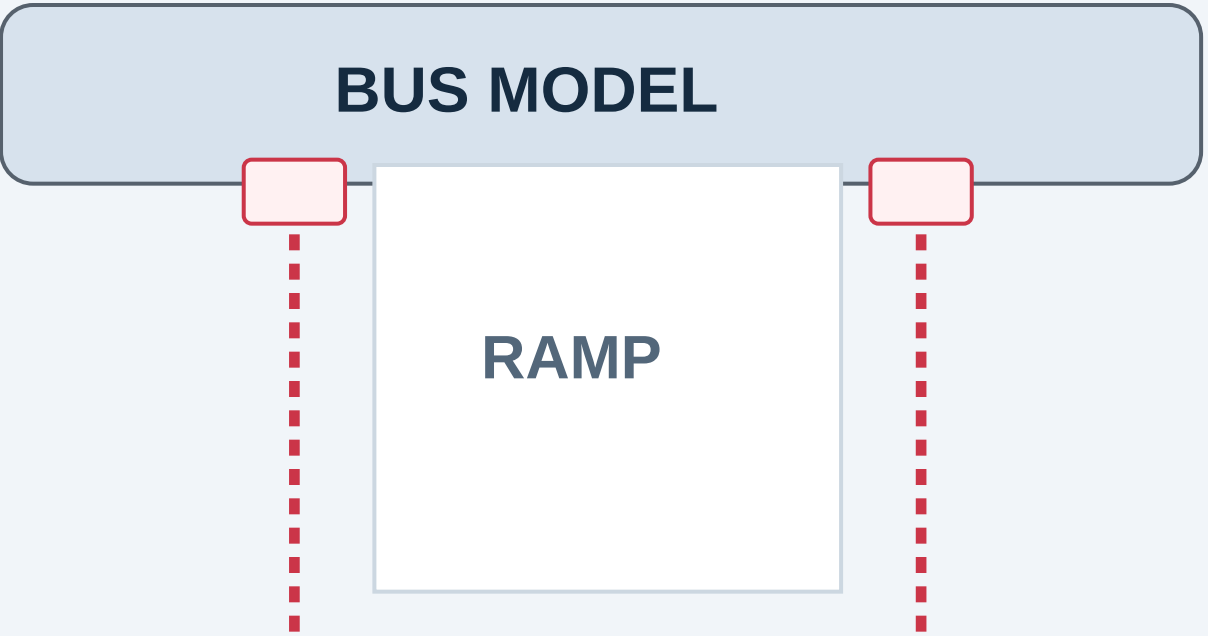
Check before switching on

Verify module voltage/polarity and the adapter G / D / S labels.
The RYS1230 name covers different optics and seller variants.
Keep the entire 5 mW beam on an enclosed matte target.

Presenting the APAS model

The laser circuit demonstrates a visible warning zone. It does not detect passengers or obstacles.

Physical arrangement



Line optics shown schematically; crosshair / dot optics differ.

Suggested presentation sequence

1. Power ESP32 by USB. Keep S1 OFF while aligning.
2. Point all projected light down into the model enclosure.
3. Switch S1 ON; firmware keeps both GPIO outputs LOW.
4. Press S2: illuminate the ramp warning area.
5. Press again: switch both laser channels OFF.

For the full APAS controller, tie lights to the ramp states.

Scope and safety

RYS1230 is an emitter, not an obstacle detector.
Section 7 still needs independent sensing and motor interlocks.
Visible 5 mW lasers can be hazardous to direct viewing.
Contain direct and reflected beams; do not aim at people.
Use a line lens for boundaries; two spots do not outline a zone.

References and design basis

APAS Design Document: Proposed Solution sections 2, 3 and 7
[Espressif: ESP32-S3-DevKitC-1 GPIO and power reference](#)
[AOS: AO3400A MOSFET datasheet \(2.5 V gate specification\)](#)
[Motorobit: RYS1230 listing \(3-5 V, <25 mA, 650 nm, 5 mW\)](#)
[FDA: Frequently Asked Questions About Lasers](#)